

Stabilities and instabilities of the 2D Euler equations on the torus

Hamiltonian splitting, symmetry decomposition, and a novel instability criterion

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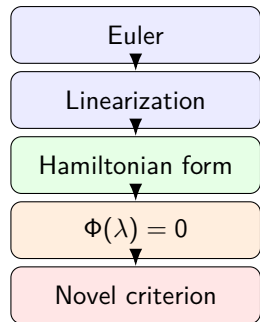
Semester Project

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Main question

How can we detect spectral instabilities of a steady state for the 2D incompressible Euler equations?

- 1 2D Euler on the torus and steady vortices.
- 2 Linearization around Taylor–Green.
- 3 Hamiltonian structure and conserved energy.
- 4 Finite dimensional instability criterion.
- 5 Application to the (1, 1) Taylor–Green vortex.
- 6 Novel contribution: instability characterization for the (2, 2) case.



2D Euler equations on the torus

Taking the curl of the Navier–Stokes equations gives the following vorticity formulation

$$\partial_t \omega + u \cdot \nabla \omega = 0 \quad \text{with } \omega = \partial_x u_2 - \partial_y u_1.$$

On $\mathbb{T}^2 = \mathbb{R}^2 / (2\pi\mathbb{Z})^2$ one writes

$$\begin{cases} \partial_t \omega + u \cdot \nabla \omega = 0, & (x, y) \in \mathbb{T}^2, \ t \geq 0, \\ u = \nabla^\perp \psi = (-\partial_y \psi, \partial_x \psi), \\ \omega = \Delta \psi. \end{cases}$$

Physical interpretation

Vorticity is transported by the flow. Since there is no viscosity, the dynamics is conservative rather than dissipative.

Historical background

A major theme in hydrodynamic stability is to understand when a steady flow keeps perturbations bounded and when it develops growing modes.

Steady solutions: from simple geometries to cellular flows

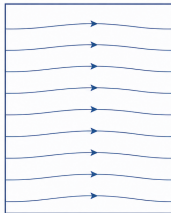
A stationary solution is obtained whenever $\partial_t \omega = 0$, or analogously

$$\partial_t \omega = -u \cdot \nabla \omega = -\{\psi_E, \omega_E\} = 0, \quad \text{where} \quad \{a, b\} := \nabla^\perp a \cdot \nabla b.$$

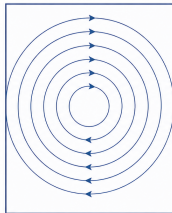
One such case is

$$\omega_E = F(\psi_E),$$

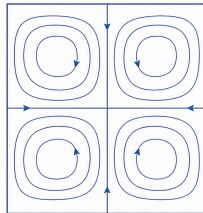
so that vorticity is constant along streamlines.



Shear-like geometry



Radial vortices



Taylor-Green geometry

The (1, 1) Taylor–Green vortex

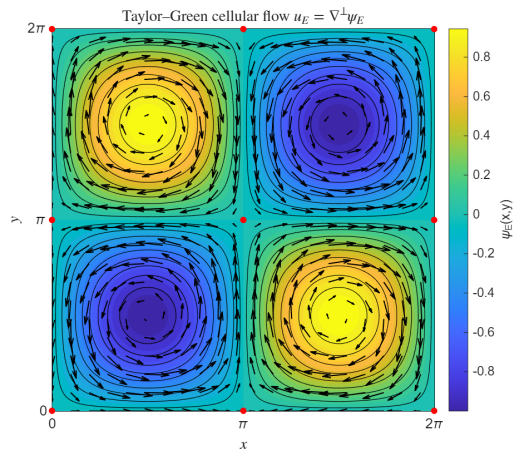
Why Taylor–Green is interesting

The (1, 1) Taylor–Green vortex is analytic, highly symmetric, and has hyperbolic stagnation points. Its spectral stability is therefore a natural test case beyond shear flows and radial vortices.

The stream function and vorticity are

$$\psi_E(x, y) = -\sin x \sin y,$$

$$\omega_E = \Delta\psi_E = 2\sin x \sin y,$$



Velocity and streamlines of the (1,1) Taylor–Green vortex.

Deriving the linearized Euler operator

Starting from

$$\begin{cases} \partial_t \omega + u \cdot \nabla \omega = 0, & (x, y) \in \mathbb{T}^2, \ t \geq 0, \\ u = \nabla^\perp \psi = (-\partial_y \psi, \partial_x \psi), & \omega = \Delta \psi. \end{cases}$$

Let

$$\omega = \omega_E + w, \quad \psi = \psi_E + \phi, \quad \Delta \phi = w.$$

we insert the perturbation and discard quadratic terms in (w, ϕ) to obtain the linearized equation:

$$\partial_t w = \mathcal{L}_E w, \quad \boxed{\mathcal{L}_E w = -\{\psi_E, (\text{Id} + 2\Delta^{-1}) w\}}.$$

Spectral stability problem

Spectral instability means that $\exists f$ s.t.

$$\mathcal{L}_E f = \lambda f, \quad \text{Re } \lambda > 0.$$

Definition (Hamiltonian operator)

Let X be a Hilbert space. A densely defined operator $\mathcal{L} = \mathcal{J}\mathcal{H}$ is Hamiltonian if

- 1 $\mathcal{J} : \text{Dom}(\mathcal{J}) \subset X \rightarrow X$ is densely defined, closed and skew-adjoint;
- 2 $\mathcal{H} : X \rightarrow X$ is bounded and self-adjoint.

For Taylor–Green,

$$\boxed{\mathcal{L}_E = \mathcal{J}\mathcal{H}}, \quad \mathcal{J} = -\{\psi_E, \cdot\}, \quad \mathcal{H} = \text{Id} + 2\Delta^{-1}.$$

The conserved energy $q_{\mathcal{H}}$

For functions that satisfy the Hamiltonian evolution

$$\partial_t f = \mathcal{J}\mathcal{H}f,$$

define

$$q_{\mathcal{H}}(f) := \langle \mathcal{H}f, f \rangle_X$$

Then

$$\frac{d}{dt} q_{\mathcal{H}}(f(t)) = \langle \mathcal{J}\mathcal{H}f, \mathcal{H}f \rangle + \langle \mathcal{H}f, \mathcal{J}\mathcal{H}f \rangle = 0,$$

because $\mathcal{J}^* = -\mathcal{J}$.

Consequence

If $\mathcal{H}$ were positive definite, then $q_{\mathcal{H}}(f) = \langle \mathcal{H}f, f \rangle = C_f > 0 \forall f$ with a strict inequality. Assume by contradiction that $f(t) = e^{\lambda t} f_0$ is an unstable eigenvector, we would have

$$q_{\mathcal{H}}(f(t)) = e^{2 \operatorname{Re} \lambda t} q_{\mathcal{H}}(f_0) = e^{2 \operatorname{Re} \lambda t} C_{f_0} \rightarrow +\infty, \quad \Rightarrow \times .$$

Negative directions and Krein signature

Let $\mathcal{L}f = \lambda f$ and $\operatorname{Re} \lambda > 0$. Then

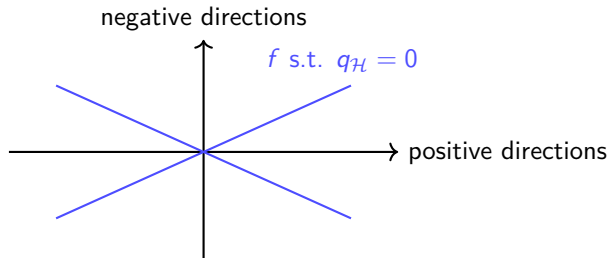
$$\lambda \langle f, \mathcal{H}f \rangle = \langle \mathcal{J}\mathcal{H}f, \mathcal{H}f \rangle = -\langle \mathcal{H}f, \mathcal{J}\mathcal{H}f \rangle = -\lambda \langle \mathcal{H}f, f \rangle,$$

so

$$q_{\mathcal{H}}(f) = \langle \mathcal{H}f, f \rangle = 0.$$

Lemma

If $\mathcal{H}$ has exactly k **negative eigenvalues**, then $\mathcal{L} = \mathcal{J}\mathcal{H}$ has at most k linearly independent **unstable eigenfunctions**.



Fourier structure of the Taylor–Green energy

For $e_{j,k} = e^{i(jx+ky)}$,

$$\Delta^{-1} e^{i(jx+ky)} = -\frac{1}{j^2 + k^2} e^{i(jx+ky)}.$$

Therefore

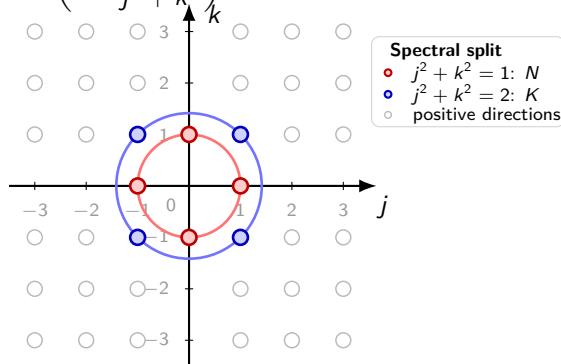
$$\mathcal{H}e_{j,k} = (\text{Id} + 2\Delta^{-1})e_{j,k} = \left(1 - \frac{2}{j^2 + k^2}\right) e_{j,k}.$$

Negative directions

$$N := \text{span}\{\sin x, \sin y, \cos x, \cos y\}.$$

Kernel directions

$$K := \text{span}\{\sin x \sin y, \sin x \cos y, \cos x \sin y, \cos x \cos y\}.$$



Theorem (Instability criterion)

Let X be a Hilbert, $\mathcal{L} = \mathcal{J}\mathcal{H}$ an Hamiltonian operator and assume that $\mathcal{H}$ admits a decomposition

$$\mathcal{H} = \mathcal{H}_s + \mathcal{H}_u$$

- such that:
- (i) $\mathcal{H}_s : X \rightarrow X$ is self-adjoint and coercive, $\inf_{\|f\|_X = 1} \langle \mathcal{H}_s f, f \rangle_X > 0$,
 - (ii) $\mathcal{H}_u : X \rightarrow V_u$ has finite rank, $n := \dim V_u < +\infty$,
 - (iii) $\mathcal{H}_s \mathcal{H}_u = \mathcal{H}_u \mathcal{H}_s$.

Define the skew-adjoint operator $\mathcal{T} := \mathcal{H}_s^{1/2} \mathcal{J} \mathcal{H}_s^{1/2}$ and set $\Phi(\lambda) := \det (\text{Id} + \mathcal{H}_u \mathcal{H}_s^{-1} (\mathcal{T} - \lambda)^{-1} \mathcal{T})$.
Then

$$\lambda \in \sigma(\mathcal{L}) \setminus i\mathbb{R} \iff \Phi(\lambda) = 0.$$

Finite-dimensional reduction theorem

For Taylor–Green, $\mathcal{H}$ is diagonal:

$$\mathcal{H} = \sum_{(j,k) \neq (0,0)} \mu_{j,k} \mathbf{e}_{j,k} \otimes \mathbf{e}_{j,k}, \quad \mu_{j,k} := 1 - \frac{2}{j^2 + k^2}.$$

Let $I_N := \{(j,k) \neq (0,0) : \mu_{j,k} \leq 0\}$. For $(j,k) \in I_N$, choose $s_{j,k}, h_{j,k} > 0$ s.t. $\mu_{j,k} = s_{j,k} - h_{j,k}$. Then

$$\mathcal{H} = \underbrace{\sum_{(j,k) \notin I_N} \mu_{j,k} \mathbf{e}_{j,k} \otimes \mathbf{e}_{j,k} + \sum_{(j,k) \in I_N} s_{j,k} \mathbf{e}_{j,k} \otimes \mathbf{e}_{j,k}}_{\mathcal{H}_s} - \underbrace{\sum_{(j,k) \in I_N} h_{j,k} \mathbf{e}_{j,k} \otimes \mathbf{e}_{j,k}}_{\mathcal{H}_u}.$$

Rank-one case

If $V_u = \text{span}\{E_u\}$ and then $\mathcal{H}_u E_u = -h_u E_u$, $\mathcal{H}_s E_u = s_u E_u$, $s_u > 0$,

$$\Phi(\lambda) = 1 - \frac{h_u}{s_u \|E_u\|_X^2} \langle (\mathcal{T} - \lambda)^{-1} \mathcal{T} E_u, E_u \rangle_X.$$

Properties of the Stability Determinant $\Phi(\lambda)$

Rank-one formula at zero

If $V_u = \text{span}\{E_u\}$ and P is the orthogonal projection on the kernel of $\mathcal{J}$, then

$$\Phi(0) = 1 + \frac{h_u}{s_u} \left(\frac{1}{s_u \|E_u\|_X^2} \langle (P\mathcal{H}_s^{-1}P)^{-1}PE_u, PE_u \rangle_X - 1 \right).$$

Monotonicity of the stability determinant

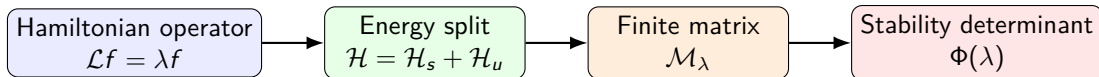
The stability determinant satisfies $\Phi(\lambda) \in \mathbb{R}$ for $\lambda \in \mathbb{R} \setminus \{0\}$, and Φ is an even function, increasing with respect to $|\lambda|$. In particular,

$$\lim_{\lambda \rightarrow +\infty} \Phi(\lambda) = 1$$

Detection of real instabilities

If $\Phi(0) < 0$ and $\Phi(\lambda) \rightarrow 1$ as $\lambda \rightarrow +\infty$, then by continuity there is a positive real unstable eigenvalue.

What did we achieve?



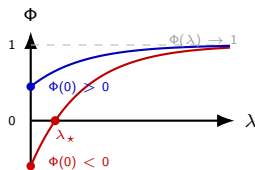
Stability Determinant $\Phi(\lambda)$:

Intability criterion

$$\Phi(\lambda) = 0$$



$$\lambda \in \sigma(\mathcal{L}) \setminus i\mathbb{R}$$



$$\Phi(0) < 0$$

one positive
real root:

$$\exists \lambda_* > 0$$

$$\Phi(0) > 0$$

no positive
real root

$$\nexists \lambda > 0$$

Non perturbative

No small parameter is followed. The reduction is exact for $\lambda \in \mathbb{C} \setminus i\mathbb{R}$.

Fixed dimension

The dimension is the number of isolated unstable energy directions, not a truncation size.

Fourier structure of the Taylor–Green transport operator

$$\psi_E(x, y) = -\sin x \sin y, \quad \mathcal{J} = -\{\psi_E, \cdot\} = \cos x \sin y \partial_y - \sin x \cos y \partial_x.$$

Parity is preserved

$$g \in A_x B_y, \quad A, B \in \{\text{Ev}, \text{Odd}\}.$$

$$\mathcal{J}g = \underbrace{\cos x}_{\text{Ev}_x} \underbrace{\sin y}_{\text{Odd}_y} \partial_y g - \underbrace{\sin x}_{\text{Odd}_x} \underbrace{\cos y}_{\text{Ev}_y} \partial_x g$$

$$: [A_x B_y] \mapsto [\text{Ev}_x \text{Odd}_y \partial_y B_y] + [\text{Odd}_x \partial_x A_x \text{Ev}_y]$$

$$: [A_x B_y] \mapsto [A_x B_y] + [A_x B_y].$$

$$\boxed{\mathcal{J}(A_x B_y) \subset A_x B_y}$$

Fourier modes are shifted by one in each variable

Take for instance

$$g(x, y) = \sin(jx) \cos(ky) \in \text{Odd}_x \text{Ev}_y.$$

$$\mathcal{J}g = -k \cos(x) \sin(jx) \sin(y) \sin(ky) - j \sin(x) \cos(jx) \cos(y) \cos(ky).$$

$$(e^{ix} \pm e^{-ix}) \overset{\text{x-variable}}{(e^{ijx} \pm e^{-ijx})} \rightsquigarrow e^{i(j+1)x}, e^{i(j-1)x}, e^{-i(j-1)x}, e^{-i(j+1)x}$$

$$(e^{iy} \pm e^{-iy}) \overset{\text{y-variable}}{(e^{iky} \pm e^{-iky})} \rightsquigarrow e^{i(k+1)y}, e^{i(k-1)y}, e^{-i(k-1)y}, e^{-i(k+1)y}$$

$$\mathcal{J}g \in \text{span} \left\{ e^{i(\pm j \pm 1)x} e^{i(\pm k \pm 1)y} \right\} = \text{span} \left\{ e^{i\alpha x} e^{i\beta y} : \alpha = +j \pm 1, \beta = +k \pm 1 \right\}.$$

$$(j, k) \mapsto (j \pm 1, k \pm 1) \quad \text{no index can stay fixed}$$

$$\mathcal{J}((A_x B_y)^{[\text{ev}]}) \subset (A_x B_y)^{[\text{ev}]}, \quad \mathcal{J}((A_x B_y)^{[\text{odd}]}) \subset (A_x B_y)^{[\text{odd}]}$$

Reason for the decomposition

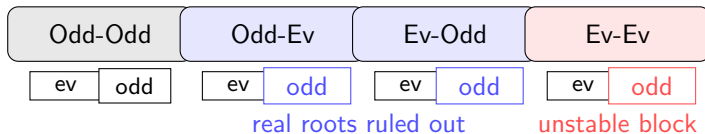
$\mathcal{H}$ is diagonal in Fourier variables, while $\mathcal{J} = -\{\psi_E, \cdot\}$ acts on a general mode (j, k) by producing the four new shifted modes $(j \pm 1, k \pm 1)$. Hence these parity classes are invariant for the full operator $\mathcal{L}_E$.

The operator preserves parity in each variable:

$$L_0^2(\mathbb{T}^2) = \text{Odd}_x \text{Odd}_y \oplus \text{Odd}_x \text{Ev}_y \oplus \text{Ev}_x \text{Odd}_y \oplus \text{Ev}_x \text{Ev}_y.$$

Each block splits again according to the parity of $j + k$ in the trigonometric basis:

$$A_x B_y = (A_x B_y)^{[\text{ev}]} \oplus (A_x B_y)^{[\text{odd}]}.$$



Stable blocks versus unstable blocks

Invariant sector	Energy information	Spectral conclusion
$(\text{Odd}_x \text{Odd}_y)^{[\text{odd}]}$ and all $[\text{ev}]$ sectors	no negative direction of $\mathcal{H}$	purely imaginary spectrum
$(\text{Odd}_x \text{Ev}_y)^{[\text{odd}]}$ and $(\text{Ev}_x \text{Odd}_y)^{[\text{odd}]}$	one negative direction	possible roots are real; criterion gives no positive real root
$(\text{Ev}_x \text{Ev}_y)^{[\text{odd}]}$	two negative directions	actual complex unstable eigenvalues occur

Idea of the proof

First, the absence of negative directions gives a sharp exclusion criterion. Relying on the monotonicity for our *stability determinant*, the value $\Phi(0)$ is then used either to exclude real roots or to locate complex roots by a winding-number argument.

Complete spectrum away from the imaginary axis

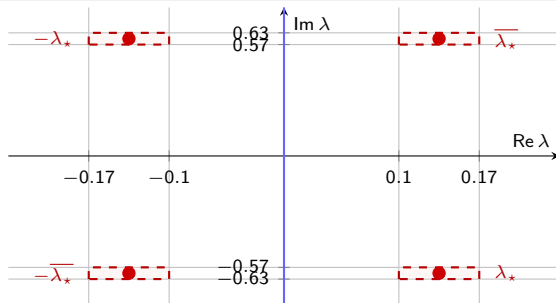
Theorem (Taylor–Green spectral instability)

Let $\mathcal{L}_E$ be the linearized Euler operator around the $(1, 1)$ Taylor–Green vortex. There exists a unique

$$\lambda_\star \in [0.10, 0.17] - i[0.57, 0.63]$$

such that

$$\sigma(\mathcal{L}_E) \setminus i\mathbb{R} = \{\lambda_\star, \bar{\lambda}_\star, -\lambda_\star, -\bar{\lambda}_\star\}.$$



From (1, 1) to (2, 2) Taylor–Green

For the generalized family

$$\psi_{m,n}(x, y) = -\sin(mx) \sin(ny),$$

the linearized operator is $\mathcal{L}_{m,n} = \mathcal{J}_{m,n} \mathcal{H}_{m,n}$,

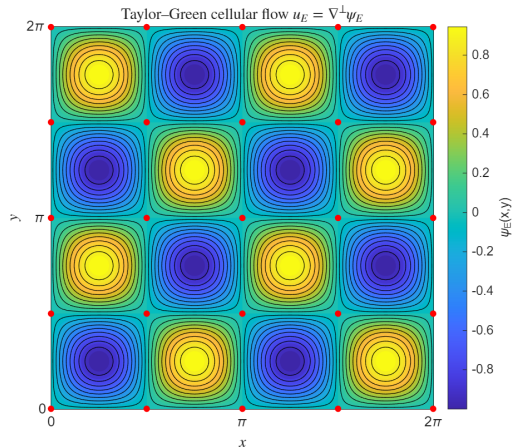
$$\mathcal{J}_{m,n} = -\{\psi_{m,n}, \cdot\}, \quad \mathcal{H}_{m,n} = \text{Id} + (m^2 + n^2) \Delta^{-1}.$$

For the (2, 2) Taylor–Green,

$$\mathcal{H}_{2,2} = \text{Id} + 8 \Delta^{-1}, \quad \mathcal{H}_{2,2} e_{j,k} = \left(1 - \frac{8}{j^2 + k^2}\right) e_{j,k}.$$

New behaviour

The (2, 2) splitting produces additional invariant blocks. In some of them the spectral properties can be recovered from the study of the (1, 1) case.



Velocity and streamlines of the (2,2) Taylor–Green vortex.

Proposition (Sign of $\Phi(0)$ in the rank-1 case)

Let $\mathcal{X}$ be a real invariant block for the Hamiltonian operator $\mathcal{L} = \mathcal{J}\mathcal{H}$, let $K := \text{Ker}\mathcal{J} \cap \mathcal{X}$ and assume that $\mathcal{H}$ is invertible and has exactly one negative direction in $\mathcal{X}$, generated by

$$\|E_u\| = 1, \quad \mathcal{H}E_u = -\omega_u E_u, \quad \omega_u > 0.$$

Define

$$\tilde{C} := \inf \{ \langle \mathcal{H}^{-1}\varphi, \varphi \rangle : \varphi \in K, \langle \varphi, E_u \rangle = 1 \}.$$

Then the value of the stability determinant at the origin satisfies

$$\text{sign } \Phi(0) = -\text{sign } \tilde{C}.$$

Proof idea

Choose the admissible rank-one splitting

$$\mathcal{H} = \mathcal{H}_s + \mathcal{H}_u, \quad \mathcal{H}_s E_u = s_u E_u, \quad \mathcal{H}_u E_u = -h_u E_u, \quad h_u = s_u + \omega_u.$$

From the formula for the limit of the stability determinant, defining $K = \text{Ker} \mathcal{J} \cap \mathcal{X}$ and $P = \text{Proj}_K$,

$$\Phi(0) = 1 + \frac{h_u}{s_u} \left(\frac{1}{s_u} \underbrace{\langle (P \mathcal{H}_s^{-1} P)^{-1} P E_u, P E_u \rangle}_{=: A} - 1 \right).$$

The term A is the only non explicit one. By the Legendre transform $A = \sup_{\varphi \in K} \{2 \langle \varphi, E_u \rangle - \langle \mathcal{H}_s^{-1} \varphi, \varphi \rangle\}$.

Write $x := \langle \varphi, E_u \rangle$, $\varphi = x\psi$, with $\langle \psi, E_u \rangle = 1$. Then $A = \sup_{\substack{x \in \mathbb{R} \\ \langle \psi, E_u \rangle = 1}} \{2x - x^2 \langle \mathcal{H}_s^{-1} \psi, \psi \rangle\}$.

Define $B := \inf_{\substack{\psi \in K \\ \langle \psi, E_u \rangle = 1}} \langle \mathcal{H}_s^{-1} \psi, \psi \rangle$. Optimizing in x , we get $A = 1/B$. Hence the determinant formula becomes

$$\Phi(0) = 1 + \frac{h_u}{s_u} \left(\frac{1}{s_u B} - 1 \right).$$

Equivalently, $B = \tilde{C} + \frac{h_u}{s_u \omega_u}$. Substituting this into the determinant formula gives

$$s_u \Phi(0) = -\frac{\omega_u}{B} \tilde{C}.$$

From the dual sign to negative energy

Lemma

Let $\mathcal{X}$, E_u , K and $\tilde{C}$ be as in the previous proposition.

Then

$$\tilde{C} > 0 \iff \exists f \in K^\perp \text{ such that } \langle \mathcal{H}f, f \rangle < 0.$$

Therefore,

$$\exists f \in K^\perp \text{ with } \langle \mathcal{H}f, f \rangle < 0 \implies \tilde{C} > 0 \implies \Phi(0) < 0.$$

Since

$$\lim_{\lambda \rightarrow +\infty} \Phi(\lambda) = 1,$$

continuity gives

$$\exists \lambda_\Delta > 0 \text{ such that } \Phi(\lambda_\Delta) = 0.$$

Interpretation: Krein signature

The quadratic form $f \mapsto \langle \mathcal{H}f, f \rangle$ is the Krein signature. The criterion says that the real block is unstable precisely when $K^\perp$ still intersects the negative cone of this signature.

Application strategy for the $(2, 2)$ real block

The abstract proposition reduces the problem to a concrete construction:

$$\text{find } f \in (\text{Ker } \mathcal{J}_{2,2})^\perp \text{ with } \langle \mathcal{H}_{2,2} f, f \rangle < 0.$$

Orthogonality to the kernel

A robust way to guarantee

$$f \in (\text{Ker } \mathcal{J}_{2,2})^\perp$$

is to write

$$f = \mathcal{J}_{2,2}^* g.$$

Since $\mathcal{J}_{2,2}$ is skew-adjoint, $\mathcal{J}_{2,2}^* = -\mathcal{J}_{2,2}$ and $\text{Ran}(\mathcal{J}_{2,2}^*) \subset (\text{Ker } \mathcal{J}_{2,2})^\perp$.

Negative energy

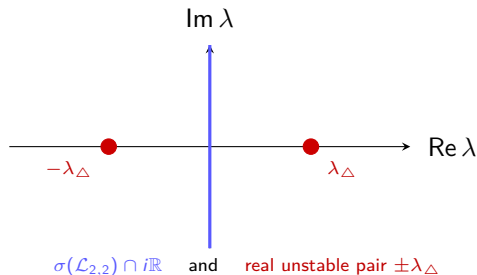
The function g is chosen so that f has a large projection onto the low Fourier mode $\cos(x + y)$, which is the negative direction in the relevant real block.

The explicit test function

The constructed function can be written as

$$\begin{aligned} f_{\Delta} = & 2 \cos(x + y) + 2 \cos(3x - y) + 2 \cos(x - 3y) \\ & - 2 \cos(5x + y) - 2 \cos(x + 5y) \\ & - \cos(5x - 3y) - \cos(3x - 5y). \end{aligned}$$

Spectrum of (2, 2) Taylor–Green on $L_{[1,1]}^{[0],c}$



Real instability in the (2,2) Taylor–Green vortex

The construction gives $f_{\Delta} \in (\text{Ker } \mathcal{J}_{2,2})^{\perp}$, $\langle \mathcal{H}_{2,2} f_{\Delta}, f_{\Delta} \rangle < 0$. By the novel proposition,

$$\tilde{C} > 0 \iff \Phi(0) < 0,$$

and therefore Φ has single a positive **real** zero.

What the novel contribution adds

Original paper

- Gives a general finite-dimensional criterion.
- Fully characterizes the unstable spectrum of $(1, 1)$ Taylor–Green.
- Explains how generalized vortices can be treated block by block.

Our novel criterion

- Reformulates the real-instability test through the sign of $\tilde{C}$.
- Proves the equivalence with the existence of a negative-energy vector in $(\text{Ker } \mathcal{J})^\perp$.
- Constructs an explicit finite Fourier test function for the $(2, 2)$ real block.

Hamiltonian signature + kernel geometry $\implies$ real eigenvalue.

Take-home message

- ① The 2D Euler equation is conservative, but steady flows may be spectrally unstable.
- ② Linearization around Taylor–Green has the Hamiltonian form

$$\mathcal{L}_E = \mathcal{J}\mathcal{H}, \quad \mathcal{J} = -\{\psi_E, \cdot\}, \quad \mathcal{H} = \text{Id} + 2\Delta^{-1}.$$

- ③ Negative directions of $\mathcal{H}$ are the only possible source of unstable eigenvalues.
- ④ The paper reduces the unstable spectrum to zeros of a finite-dimensional determinant Φ .
- ⑤ The novel contribution turns the real $(2, 2)$ block into a concrete negative-energy construction:

$$\exists f \in (\text{Ker}\mathcal{J})^\perp, \quad \langle \mathcal{H}f, f \rangle < 0.$$

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